Carol Genetti. 2018. *How languages work: An introduction to language and lingustics. Cambridge University Press.*

2 Phonetics

Physical Dimensions of Speech Sounds

KEY TERMS

- Subglottal system
- Voicing
- Voiced vs. voiceless consonant
- Orthography
- Fundamental frequency vs. pitch
- Supralaryngeal vocal tract and its subparts (lips, alveolar ridge, etc.)
- Places of articulation (bilabial, labiodental, etc.)
- Manners of articulation (stop, fricative, etc.)
- Obstruent vs. sonorant
- International Phonetic Alphabet (IPA)
- Vowel
- Co-articulation
- Suprasegmental

CHAPTER PREVIEW

Phonetics is the branch of linguistics that is concerned with the scientific study of speech sounds. The study of phonetics can provide answers to many questions that you might have wondered about at one time or another. For example, what does it mean to say that someone has a higher-pitched voice than someone else? What makes a tone language like Mandarin Chinese different from a non-tonal language such as English or Spanish? How do English pairs of words such as the verb *import* and the noun *import* differ?

Several areas of phonetics have been the focus of research into the features of speech. **Articulatory phonetics** is concerned with how the vocal organs produce speech. **Acoustic phonetics** deals with the physical characteristics of speech, such as the duration, frequency, and intensity of sounds. **Auditory phonetics** examines the perception of speech by the auditory system. Acoustic, articulatory, and auditory phonetics are all interrelated, since changing the articulatory configuration of the vocal tract results in acoustic changes which in turn potentially influence the perception of a sound. In this chapter, we will consider the first of these areas of phonetic research, providing an overview of the field as well as answers to the questions posed above. Students will be introduced to the tasks of discerning different speech sounds, describing them in phonetic terms, and accurately recording them using the International Phonetic Alphabet.

LIST OF AIMS

At the end of this unit, students will be able to:

- **identify the parts of the vocal tract responsible for producing different sounds;**
- **describe the manner and places of articulation of consonants and vowels;**
- **produce the phonetic symbols for English sounds;**
- **transcribe English words using the IPA;**
- **read English words and passages written in the IPA;**
- **use the IPA chart as a reference for sounds in languages other than English.**

2.1 The Speech Organs

The physical production of speech requires intricate coordination between several parts of the upper body, from the stomach all the way up to the nose. It is common to divide the speech organs into three subsystems (see Figure 2.1): the subglottal system, the larynx, and the supralaryngeal (or supraglottal) system.

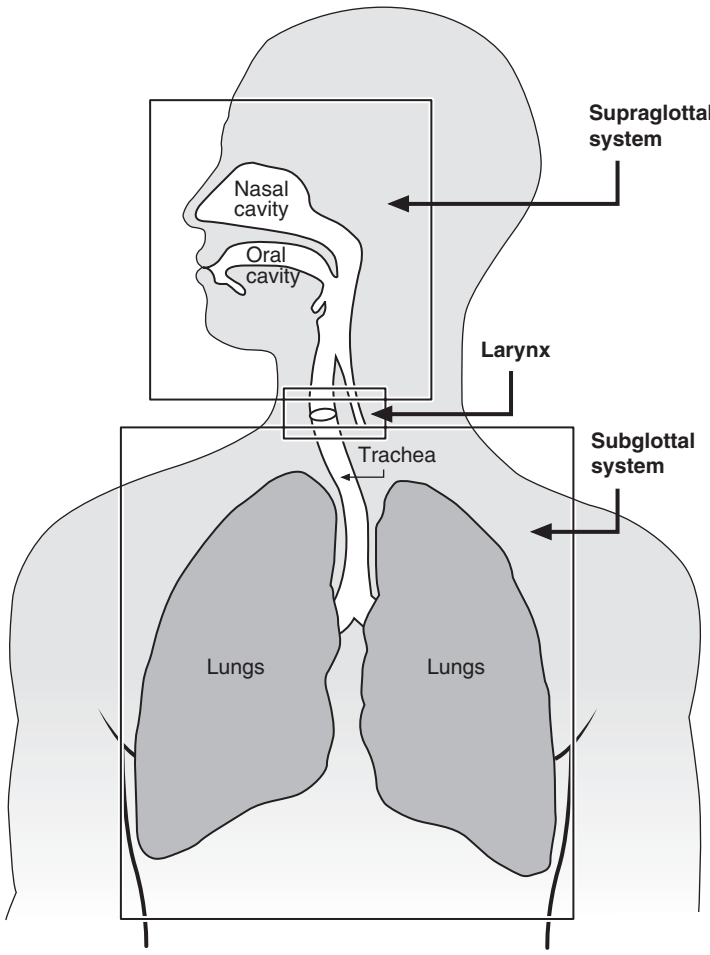


Figure 2.1 Three subsystems of speech articulation

2.1.1 The Subglottal System

The **subglottal system** includes the lungs and the trachea (or windpipe), which provide the air that the upstream articulators manipulate to produce sound. The lungs function like balloons, recoiling after inspiration and setting the air molecules in the vocal tract in motion.

2.1.2 The Larynx

Moving up from the lungs and trachea, the **larynx** is the source for many of the sounds produced in speech. It is located behind the thyroid cartilage (or Adam's apple), which is the bump you can feel on the front of the neck if you lean your head back. ***The larynx contains two vocal folds that vibrate during voiced sounds such as z or v***. To feel the vocal folds vibrate, try placing your fingers on the thyroid cartilage in the front of your neck while making a prolonged [zzzzzz] sound. You will feel the cartilage vibrating. Compare this to what happens when you make a long [ssssss] sound. There is no vibration. Vocal fold vibration, otherwise known as **voicing**, does not require any active motion beyond positioning the vocal folds close enough together that the passage of air between them causes them to vibrate. As long as the air pressure below the larynx is less than the pressure above the larynx, you can sustain a voiced sound. When you produce the sound [zzzzz], you are producing a **voiced consonant**.

As you have already learned when you made the [ssssss] sound, it is also possible to make the same sounds without vocal fold vibration, in which case, you produce a **voiceless consonant**. For example, by turning off voicing in *z* you get *s* and by turning off voicing during *v* you get *f*. (See Sidebar 2.3.) Physically, devoicing of these sounds is achieved by opening the larynx wider than for their voiced counterparts.

In English there are two sounds that only involve the larynx and not any articulators above the larynx. One is the *h* sound in words like *hat* or *ahead*. The other is the glottal stop found in the middle of the expression *uh-oh*. Try saying *uh-oh* emphatically while your hand is on your larynx and your head is leaned back. You will feel an abrupt

SIDEBAR 2.1

You can find definitions for key terms and bolded terms throughout this chapter in the Glossary (at the back of this book or on the student resources website). Additional online resources for this chapter include a study guide, accompanying audio files, a review quiz, an articulators quiz, vocabulary quizzes, IPA flashcards, an interactive IPA chart with audio, and a phonetic transcription exercise.

SIDEBAR 2.2

For a description of some of the ways people can modify a normal speaking voice, most often with effects produced in the larynx, see Section 10.3.

Note: Throughout the rest of the book, sidebars like this one will cross-reference sections or textboxes in other chapters or language profiles. For example, "Section 10.3" refers to the third section in Chapter 10 (Prosody). Other sidebars might mention a section number that starts with "LP" (Language Profile), e.g., "Section LP8.1" refers to the first section of the Bardi Language Profile (LP8), and "Textbox LP8.4" refers to the fourth textbox in the Bardi Language Profile.

SIDEBAR 2.3

In phonetics (and other linguistic fields) it is important to distinguish between how a sound is spelled, the **orthography**, and how it is phonetically transcribed. In this chapter, we will initially use italics to represent words and sounds orthographically. Phonetic transcription, which is written between square brackets, is introduced later in the chapter.

stoppage of voicing during the glottal stop between the two vowels as the vocal folds come together to block off all airflow through the larynx. This is called a **glottal stop**, as you stop the airflow by closing the **glottis** (the space between the vocal folds).

By adjusting the tension of the vocal folds during voicing, you can change the fundamental frequency, and hence the pitch, of a sound. The fundamental frequency of a sound is commonly referred to as pitch, though the two terms are technically not synonymous. **Fundamental frequency** refers to the physical property of rate of vocal fold vibration, whereas **pitch** refers to the perception of the sound on a scale of low to high. Increasing fundamental frequency also typically increases the pitch. Someone with a relatively high-pitched voice thus has a relatively high fundamental frequency or a fast rate of vocal fold vibration. Conversely, someone with a low-pitched voiced has a relatively slow rate of vocal fold vibration. Try making the sound [ahhhhhh]. Now try raising the pitch of the sound while your fingers are on your larynx. You will feel the larynx tense up and rise. Now try lowering the pitch of the sound. You will feel the larynx relax and lower. In this way, you can see how an individual can easily change the fundamental frequency of his or her voice.

2.1.3 The Supralaryngeal Vocal Tract and Place of Articulation

Above the larynx is the **supralaryngeal vocal tract**, which contains most of the structures that are manipulated in speech. The articulators of the supralaryngeal vocal tract are shown in Figure 2.2. When you use different articulators to produce speech sounds, you are changing the **place of articulation** of the sound.

In discussing different places of articulation, it is useful to move from the front to the back of the mouth (see Figure 2.2), starting with the most visible organs. The lips play an important role in producing many sounds, including *p, b, m, w, f,* and *v* in English. Sounds that involve a narrowing or a complete closure of the upper and lower lip are called **bilabials**. The bilabial sounds of English include *p, b, m*, and *w*. Sounds involving the upper teeth and the lower lip are referred to as **labiodentals**. These include *f* and *v*. For labiodentals, the lower lip is the **active articulator**, since it moves to meet the upper teeth. The upper teeth are thus the **passive articulator**, since they are stationary. Most consonant articulations involve both an active and a passive articulator. As we will see, for most consonants, the tongue is the active articulator, while the upper surface of the mouth is the passive articulator.

The structures just behind the lips are relatively immobile compared to the lips. These rigid structures include the teeth, the **alveolar ridge** (the hard ridge just behind the teeth before the upper surface of the mouth becomes more domed in shape), and the **hard palate** (the domed part of the roof of the mouth). The teeth are involved in the production of the English *th* sounds in the words *think* and *this*. These sounds are produced by either sticking the tip of the tongue between the upper and lower teeth, in which case the sounds are said to be **interdental**, or placing the tip of the tongue against the back of the upper teeth, in which case the sounds are simply called **dentals**. Try saying *think* with an interdental *th* and then with a dental *th*. In order to make the comparison fair, be careful that you are not completely blocking air from leaving the mouth when making the dental *th*.

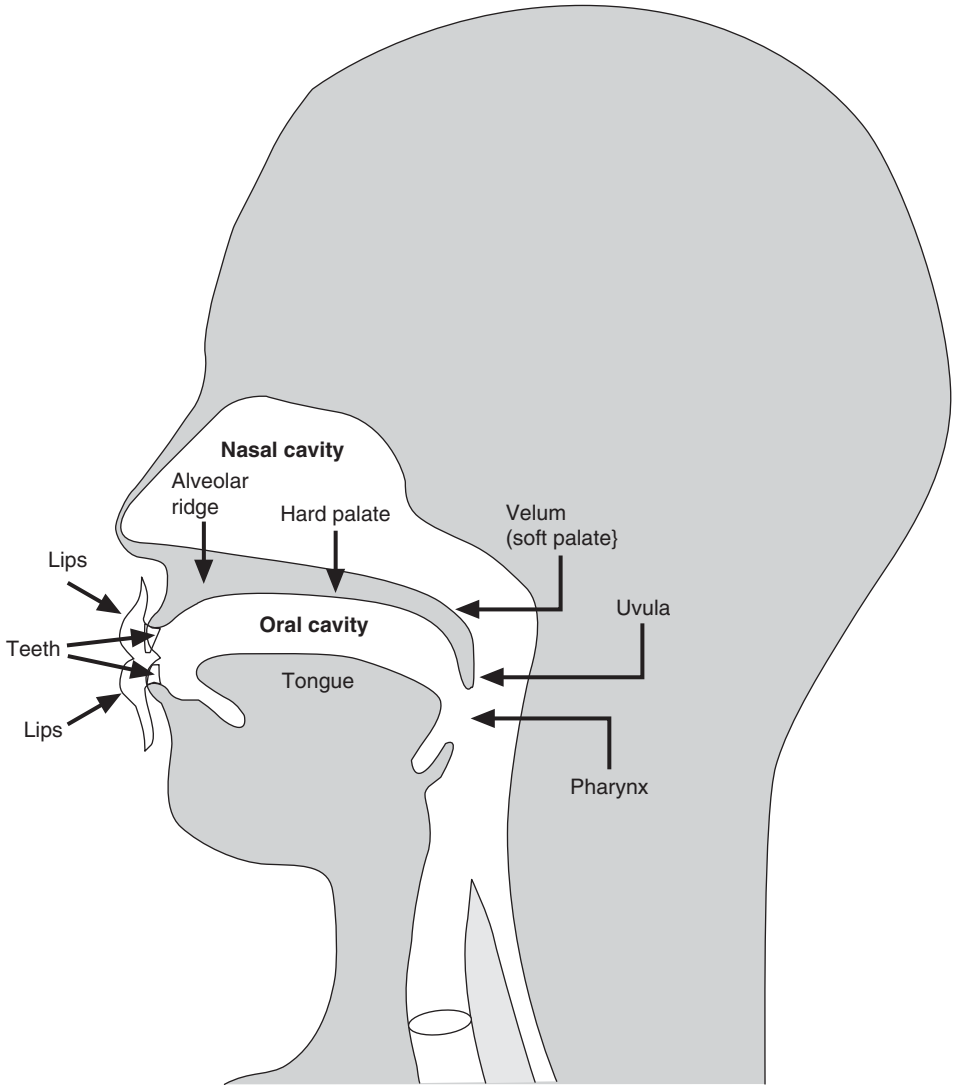


Figure 2.2 The supralaryngeal vocal tract

You will not notice much of a difference in the sound from making this small articulatory adjustment.

You may notice that the first sounds of *think* and *this* are different even though they are both spelled as *th*. The two versions of *th* differ in voicing, just as *z* and *s* were shown earlier to do: the *th* in *think* is voiceless, whereas the *th* in *this* is voiced. We will see shortly that there is a system for transcribing speech in which the voiceless and the voiced *th* are represented differently.



STOP AND REFLECT 2.1 **TONGUE POSITION AND ALVEOLAR SOUNDS**

You may note that not all of the **alveolar** sounds have the exact same point of contact for your tongue. This type of variation is quite common; in particular, *l* often involves quite a bit of contact with the back of the upper teeth, at least for many speakers of American English. Try making an extended [llll] sound and feel where the tip of the tongue is contacting the upper surface of the mouth. Now try making a [nnnnnn] sound and feel where the contact is. Does it occur at the same place as for the [llllll] sound? Even if there is a difference, you should be aware that it is common for *t, d, n*, and *l* to all be treated as alveolars in discussing the sounds of English.

Just behind the teeth is the alveolar ridge, which is the contact point between the tongue and the roof of the mouth for several sounds, including *t* (a voiceless sound), *d* (voiced), *s* (voiceless), z (voiced), *n* (voiced), and *l* (voiced). Another sound *r* (voiced) is also typically assigned to this same group, since *r* is produced with a narrowing in the vocal tract below the alveolar ridge, even though the tongue may be raised only slightly toward the roof of the mouth without touching it. See Stop and Reflect 2.1 for more discussion of alveolar sounds.

There are also sounds that are produced with the tongue contacting the area just behind the alveolar ridge. These are the **postalveolar** (or **palato-alveolar**) sounds, which include the *sh* sound in *ship* (which is a voiceless postalveolar) and the last sound in *rouge* (which is a voiced postalveolar), as well as the first sounds in *jug* (voiced) and *chug* (voiceless). To understand the relationship between alveolars and postalveolars, make an *sss* sound and then switch to a *shh* sound. You will feel the tongue sliding backwards along the upper surface of the mouth as it moves from an alveolar to a postalveolar place of articulation.

Sounds involving contact with the roof of the mouth in the center of the hard palate are simply termed **palatals**. English has a single palatal sound: the *y* sound in words like *yellow* and *young*.

Now drag your tongue backwards from the hard palate. You will notice that the upper surface of the mouth becomes softer. This area of the mouth is called the **soft palate**, or **velum**. Sounds produced by contacting the tongue and the soft palate are termed **velars**. These include the *k* (a voiceless velar sound) in words like *cat, bucket*, and *crib*, the *g* (a voiced velar) in words like *gas, go*, and *bag*, and the final *ng* sound (voiced) in words like *sing* and *lung*. The sound *w* also involves some raising of the back of the tongue toward the soft palate in addition to rounding of the lips; for this reason *w* is often labeled a labial-velar.

The soft palate, or velum, is important for distinguishing sounds involving airflow through the nose and those lacking nasal airflow. Try looking in the mirror while saying *ah* and you will see the velum rise. This raising of the velum ensures that no air escapes through the nose while you are producing the vowel. To see this, hold your finger under your nose while saying *ah*. Now try lowering the velum, holding the same tongue position for *ah* while your finger is still in place under your nose. You will feel air passing through your nose and will hear a nasal-sounding *ahn*, which is found in many languages of the world, such as French.

English also has **nasal** sounds, but they are consonants rather than vowels. Place your finger under your nose while you are making the sounds *m, n*, and the *ng* sound in *sing* and *lung*. You will feel air passing through your nose. Nasality is a separate dimension from place of articulation, since sounds can have the same place of articulation but differ in whether they are nasal or non-nasal (i.e., **oral**). The sounds *m* and *b* differ in nasality; both are voiced and both are bilabial, but only *m* is nasal. Try saying an [m] with your finger under your nose and then immediately switch to a [b]. You will feel airflow through the nose during the [m] but not during the [b] even though the lips remain closed throughout both sounds. The sounds *n* and *d* also differ only in nasality; *n* is nasal and *d* is oral. Similarly, *ng* and *g* differ in nasality; *ng* is the nasal member of the pair. Note that it is common to omit the term oral when describing oral sounds, since oral is assumed to be the default case. (See also Stop and Reflect 2.2 for a brief discussion of nasals and voicing.)



STOP AND REFLECT 2.2 **NASAL SOUNDS**

In most languages all nasals are voiced, since it is difficult to produce a voiceless nasal that is clearly audible. Try making an [m] sound and then turning off vocal fold vibration. You will wind up with a voiceless nasal, which sounds identical to the sound made when breathing through your nose.

Practice producing voiced and voiceless nasals and listen to how they sound. You will notice that it is much harder to hear a voiceless nasal than a voiced nasal. Although they are rare in the world's languages, voiceless nasals are found as consonants in some languages, such as Burmese.



Audio recordings of voiceless nasals in Burmese

2.2 Manner of Articulation

Thus far we have discussed three dimensions relevant for describing speech sounds: the voicing dimension (voiced vs. voiceless), the place of articulation dimension, and the nasality dimension (nasal vs. oral). There is one other dimension that we must consider: the narrowness of the constriction in the vocal tract. Differences in constriction narrowness are referred to as differences in **manner of articulation**.

Some sounds involve a complete closure of the vocal tract. These are called **stops**. English stops include *p, b, m, t, d, n, k, g*, and the *ng* sound. Of these sounds, *p, b, t, d, k*, and *g* are oral stops since there is no nasal airflow, while *m, n*, and *ng* are nasal stops. All stops involve two phases: a closure phase, during which the airflow through the mouth is completely blocked, and a release phase, when the constriction is released. To see this, produce just the closure for the voiceless stop *t* without releasing the tongue from the alveolar ridge. You will notice that there is complete silence, since there is no voicing. This means that voiceless stops are only identifiable through their release, which provides crucial information about place of articulation. There is another type of sound found in certain varieties of English, including for most speakers of American and Australian English, that resembles an alveolar stop in that it is produced with a complete closure at the alveolar ridge. This sound is called a **flap** and occurs in the middle of words like *pity, butter, lady*, and *ladder*. The key difference between a stop and a flap (sometimes also referred to as a **tap**) is the extreme

shortness of the closure for the flap. The tongue briefly taps the roof of the mouth before rapidly returning to position for the following sound.

It is also possible to produce sounds in which the two articulators are close together, but not so tightly occluded that no air can escape through the mouth. Sounds produced by a tight narrowing of articulators are termed **fricatives**. Fricatives are characterized by turbulence created through the random collision of air molecules either at the constriction location or, in the case of alveolar or postalveolar fricatives, by funneling air to hit the back of the teeth. Try making an emphatic [ssssss] sound while holding your hand palm down against your chin. You will feel air striking the back of your hand because the air is being directed downward after it hits the back of your upper teeth. You will also feel air striking your hand if you produce a prolonged and emphatic *sh* sound. Now try making a [hhhhhh] sound while holding your hand in the same position. You will not feel any air striking your hand. This is because the noise in [hhhhhh] is being produced directly in the larynx and not by directing air against the back of the teeth. You will also notice that the noise of *h* is much quieter than that associated with *s* or *sh*. Sounds like *s* and *sh*, which involve funneling of air against the back of the teeth are particularly noisy. These sounds are called **stridents**.

There are two sounds in English that are produced by combining a stop with a following fricative in rapid succession. These are the **affricates**, which include the *j* sound in *jug* (which is voiced) and the *ch* sounds word-initially and word-finally in *church* (which are voiceless). Since affricates have a stop phase, they are often grouped together with other stops.

Additionally, ***sounds can be produced through a slight narrowing of the vocal tract, but not enough to cause noise or a complete obstruction***. These sounds are called **approximants**. Approximants in English include the *y* sound in *yellow* and the *w* sound in *water*. Also included in the class of approximants in English is the *r* sound in words like *red, brick*, and *car*.

A final approximant to consider is the sound *l*, which resembles *t* and *d* in involving a complete closure in the middle of the alveolar ridge. There is, however, a crucial difference between *l* and these other sounds (besides the fact that *l* differs from *t* in being voiced). ***The l sound is produced with a closure only in the center of the mouth***. At least one side of the tongue (if not both, depending on the speaker) is pulled down slightly, away from the roof of the mouth. The lowering of the side part of the tongue allows air to escape the mouth, whereas the true stops *t* and *d* have a complete closure around the upper surface of the mouth. Try making an [llllll] sound and put your hand first on the left side of the mouth and then on the right side. Is the air escaping from just one or both sides of the mouth? The sound *l* has a **lateral** articulation in opposition to all of the other sounds of English, which have **central** articulations. Note that it is common to omit the term central when describing central sounds, since central is assumed to be the default case.

There are a couple of additional useful terms for grouping together certain types of sounds. One of these is the term **liquids**, which includes lateral approximants and *r*-type sounds. Another common descriptor is the term **obstruent**, which refers to the combined

set of oral stops and fricatives. Sounds that are not obstruents are the **sonorants**, which include the nasals and all of the approximants, both lateral and central.

It is important to recognize that our discussion of places and manner of articulation has focused on sounds occurring in English. In reality, there are many more sounds found in languages other than English, as you know already if you speak or have studied other languages. Some of these sounds are discussed below.

2.3 The International Phonetic Alphabet

Up to now, we have referred to the different sounds of English using the traditional symbols used in English spelling. While this strategy has worked for the most part, ***there are some limitations of using spelling (orthographic) characters to represent sounds***. One problem we have already encountered concerns the English letters *th*, which can represent either a voiceless dental fricative as in *think* or a voiced dental fricative as in *this*. Another issue is the use of two letters to represent a single sound in English. For example, the voiced velar nasal at the end of *sing* is represented by the combination *ng*. Similarly, the voiceless postalveolar fricative is written as the sequence *sh*. While it is possible to use two letters to represent a single sound, it is more efficient to use a single symbol to represent a sound that behaves phonetically as a single entity. Furthermore, there is the potential for confusion between a single phonetic sound written with two letters and a sequence of two phonetic sounds also written with two letters. For example, if one sees *ng*, how can one be sure without listening to the word, whether *ng* refers to a single voiced velar nasal, or the phonetic sequence *n* (voiced alveolar nasal) plus *g* (voiced velar oral stop), as occurs in a careful pronunciation of the compound *rain gauge*?

There is an even more serious problem with the use of English spelling to represent phonetic sounds. Many individual English letters or combinations of letters represent multiple phonetic sounds depending on the particular word. For example, the letter *x* can represent either a voiced alveolar fricative, *z*, as in *xylophone*, or the phonetic sequence of voiceless velar stop, *k*, plus voiceless alveolar fricative, *s*, as in *ox*. Similarly, the letter *o* has three different qualities in the words *ton, pond*, and *drone*. Using orthography to represent phonetic pronunciation is thus bound to cause confusion.

SIDEBAR 2.4
For a more detailed discussion of orthographic systems, see the Indonesian Language Profile, Section LP12.4.1 and Textbox LP12.4.

Fortunately, this problem is remedied by the existence of a special phonetic alphabet designed to reflect pronunciation: the International Phonetic Alphabet, abbreviated IPA. ***A crucial principle guiding the International Phonetic Alphabet is its universal one-to-one correspondence between symbols and pronunciation***. Thus, whenever you see an IPA symbol, you can be sure of its pronunciation, regardless of the language being transcribed. For example, whenever you see an IPA *m*, it will refer to a voiced bilabial nasal stop in any language. The IPA thus provides a useful tool for linguists who are transcribing words and who plan to share these transcriptions with other researchers. It is not the case, however, that all linguistic data are transcribed in IPA at all times; see Textbox 2.1.

TEXTBOX 2.1 **THE IPA AND REGIONAL TRANSCRIPTION PRACTICES**

As noted, the IPA has been designed to transcribe the sounds of every spoken language. However, it is not the case that all linguists use the IPA at all times. In many parts of the world, the transcription practices of linguists reflect the orthographic practices of the local region. In some cases, particular orthographic practices are widespread in a given region, such as the use of the symbol *š* in the transcription of North American Indian languages for the sound written with "sh" in English spelling and transcribed as [ʃ] in IPA.

In this volume, which takes data from many sources – historical and contemporary – from all over the world, the transcription system of the original source is used. Interpreting transcriptions requires careful attention; transcription notes are provided throughout the book to aid readers in this task. Learning these different systems, and how to interpret a variety of transcription practices, is part of the task of learning linguistics.

As it happens, the IPA bears close resemblance to English orthography in many respects. There are only a few major points of departure between the two systems. First, the IPA symbol for a voiceless (inter)dental fricative (as in *think*) is [θ] (IPA symbols will henceforth be written in brackets), while the symbol for its voiced counterpart (as in *this*) is [ð]. The IPA symbol for a voiceless postalveolar fricative (as in *ship*) is [ʃ], while the symbol for its voiced counterpart (as in *rouge*) is [ʒ]. The IPA represents the voiceless postalveolar affricate (as in *church*) as [tʃ] and the voiced postalveolar affricate (as in *jug*) as [dʒ]. The IPA symbol for a voiced velar nasal stop (as in *sing*) is [ŋ]. The IPA symbol for glottal stop (as in *uh-oh*) looks like a question mark but with a horizontal base rather than a period [ʔ]. In addition, the IPA symbol for a voiced palatal approximant (as in *young*) is [j], while the symbol for a central alveolar approximant (as in *red*) is an upside-down [ɹ]. Finally, the flap occurring in American English in the middle of words like *city* and *buddy* is represented with the symbol [ɾ].

SIDEBAR 2.5

See also the Nuuchahnulth Language Profile, Textbox LP5.1, about transcription conventions and the IPA.

SIDEBAR 2.6

The full chart of consonants and vowels in the International Phonetic Alphabet can be found at the back of this book. This will be a useful reference as you explore the wide variety of languages discussed in the book. An interactive IPA chart, including audio, is also available on the student resources page of the website.



Tools > Interactive IPA Chart

We are now ready to see the entire list of IPA consonant symbols relevant for describing English in Table 2.1.

In the chart, places of articulation appear as columns across the top of the chart, while manner, nasality, and lateralness are captured in rows. Sounds differing only in voicing are adjacent with the voiceless sound on the left and the voiced counterpart on the right. There are many possible combinations where no consonant is represented; this reflects the lack of such sounds in English, but does not mean that they are unattested in other languages. Note that [w] appears in both the bilabial and the velar columns, since, as we have seen, it involves constrictions simultaneously at the lips and at the velum. The flap [ɾ] appears in parentheses reflecting the fact that it is present for only certain varieties of English, such as American and Australian English.

TABLE 2.1 IPA chart for English consonants

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Palatal	Velar	Glottal
Oral stops	p b			t d			k g	ʔ
Affricates					tʃ dʒ			
Nasal stops	m			n			ŋ	
Flap (tap)				(ɾ)				
Fricatives		f v	θ ð	s z	ʃ ʒ			h
Central approximants	w			ɹ		j	w	
Lateral approximants				l				

LL: End here for Phonetics I: Consonants. Continue below for Phonetics II: Vowels.

There are also IPA symbols for vowels. ***Vowels fundamentally differ from consonants in being produced with a relatively open vocal tract***, though of course there is some movement of the tongue necessary to make different vowel sounds. The IPA chart for American English appears in Table 2.2 followed, in Table 2.3, by its counterpart for the variety of British English commonly referred to as BBC English.

STOP AND REFLECT 2.3 **VOWELS IN ENGLISH VARIETIES**

Recordings of the words in Tables 2.4 and 2.5 can be found on the *How Languages Work* website. For consistency, all transcriptions in this chapter will be based on recordings from these, as are the exercises and online resources. You should learn to hear the distinctions these speakers make and transcribe them accurately.

Accompanying Sound files for Chapter 02: Tables 2.4 and 2.5.

Your pronunciation of English might be different from these in some respects. Try pronouncing the words in Tables 2.4 and 2.5. Are your vowels the same as these or different? If different, you can find the full IPA chart online and determine which vowels you have in your own speech (your instructor can help you with this). Textboxes 2.2 and 2.3 discuss some of the variation in vowels found across English dialects.

As the tables show, the dimensions used to describe vowels differ from those used to classify consonants. ***Vowels can be described in terms of three core dimensions: backness, height, and tenseness***. In addition, vowels can differ in whether they are produced with lip rounding, as with the vowel in *boat*, or not, as with the vowel in *beet*.

There are three degrees of **height** in English: high, mid, and low. Example words illustrating the vowels of American and British English are shown in Tables 2.4 and 2.5, respectively. The mid-central vowel [ə], also known as **schwa**, is confined to unstressed syllables in English. It is pronounced with a higher tongue position than the mid-central vowel [ʌ], which is only found in stressed syllables (you can hear both of these vowels in the word *above*; the first is [ə] and the second is [ʌ]).

TABLE 2.2 IPA chart for American English vowels

		Front	Central	Back
High	Tense	i		u
	Lax	ɪ		ʊ
Mid	Tense	e		o
	Lax	ɛ	ə	
			ʌ	
Low	Lax	æ	a	

LL: the placement of [ʌ] is wrong (in both charts). I will elaborate in class.

TABLE 2.3 IPA chart for British English vowels

		Front	Central	Back
High	Tense	i		u
	Lax	ɪ		ʊ
Mid	Tense	e	ɜ	o
	Lax	ɛ	ə	ɔ
			ʌ	
Low	Lax	æ		ɑ ɒ

TEXTBOX 2.2 [ɔ] AND [a] IN AMERICAN ENGLISH

Many speakers of American English, including most from California, lack the vowel [ɔ]. For speakers who do not have [ɔ], it is because it has merged with the low vowel [a]. Speakers who have undergone this merger typically (at least in California) produce a low central vowel in words like *caught, dawn*, and *law*. Speakers who have a contrast between [ɔ] and [a] have pairs of words differing only in the vowel, e.g., *caught* with [ɔ] versus *cot* with [a] and *dawn* with [ɔ] versus *don* with [a]. Try asking several Americans to say the words *cot* and *caught*. Do any of them have a different vowel in the two words? Now try asking a speaker from Great Britain how they pronounce these words. How does their pronunciation compare with yours?

Sound files for low vowels in American English: *caught/cot*

Vowels can also be described in terms of **backness**. The front vowels of English are [i, ɪ, e, ɛ, æ]. The back vowels of English (both dialects combined) are [u, ʊ, o, ɔ, ɑ, ɒ], while the central vowels are [ə, ɜ, ʌ, a]. Several of the back and central vowels vary considerably between dialects and between speakers in their pronunciation (Textbook 2.2). The vowel [a] for American English speakers varies widely in its backness from speaker to speaker, so you may note that you have a somewhat backer pronunciation than central [a]. British English

TABLE 2.4 Example words illustrating the vowels of American English

Vowel height	Vowel	Description	English words
High	[i]	High, front, tense, unrounded	*beet, bleed, see*
	[ɪ]	High, front, lax, unrounded	*sit, fin, lip*
	[u]	High, back, tense, rounded	*boot, mood, soon*
	[ʊ]	High, back, lax, rounded	*foot, could, hood*
Mid	[e]	Mid, front, tense, unrounded	*late, rain, paid*
	[ɛ]	Mid, front, lax, unrounded	*red, send, peck*
	[ə]	Mid, central, lax, unrounded	*about, ago*
	[ʌ]	Lower-mid, central, lax, unrounded	*rut, mud, up*
	[o]	Mid, back, tense, rounded	*mode, loan, sew*
Low	[a]	Low, central, lax, unrounded	*hot, mop, rock*
	[æ]	Low, front, lax, unrounded	*cat, man, trap*

LL: depending on your speech variety, these words may not work to illustrate the vowels - we will discuss more in class

LL: the description of [ʌ] is wrong.

TABLE 2.5 Example words illustrating the vowels of British English

Vowel height	Vowel	Description	English words
High	[i]	High, front, tense, unrounded	*beet, bleed, see*
	[ɪ]	High, front, lax, unrounded	*sit, fin, lip*
	[u]	High, back, tense, rounded	*boot, mood, soon*
	[ʊ]	High, back, lax, rounded	*foot, could, hood*
Mid	[e]	Mid, front, tense, unrounded	*late, rain, paid*
	[ɛ]	Mid, front, lax, unrounded	*red, send, peck*
	[ɜ]	Mid, central, tense, unrounded	*purr, learn, sir*
	[ə]	Mid, central, lax, unrounded	*about, ago*
	[ʌ]	Lower-mid, central, lax, unrounded	*rut, mud, up*
	[o]	Mid, back, tense, rounded	*mode, loan, sew*
	[ɔ]	Mid, back, lax, rounded	*dawn, bought, north*
	[ɒ]	Low, back, lax, rounded	*hot, mop, rock*
	[ɑ]	Low, back, lax, unrounded	*half, father, hard*
Low	[æ]	Low, front, lax, unrounded	*cat, man, trap*

has a low back unrounded [ɑ] that corresponds to American English [æ] in words like *half* and to American English [a] in words like *father*. British English also has a low back rounded [ɒ] that corresponds to American English [a] in words like *hot* and *pot*. And British English has an additional central vowel in words like *purr* and *learn* that corresponds to a syllabic [ɻ] or rhotacized (i.e., r-colored) schwa [ɚ] (see Textbox 2.3) in American English.

Vowels can also differ in terms of their **tenseness**. There are several pairs of vowels in English differing only in tenseness. For example, the pair of vowels [i] as in *seat*, and [ɪ] as *sit* differ along this dimension; [i] is a tense vowel and [ɪ] is a lax vowel. Similarly, [e] is tense and [ɛ] is lax, [u] is tense and [ʊ] is lax, [o] is tense and [ɔ] is lax, and [ɜ] is tense and [ə] is lax. Tense vowels in English are longer than their lax counterparts and, for front and back vowels, also typically have a slightly more peripheral tongue position (i.e., higher and fronter in the case of front vowels, and higher and backer in the case of back vowels). You can verify this by producing an [i] sound and then changing it to an [ɪ]. You will feel the tongue lowering slightly and retracting. Now try the same exercise by switching from [e] to [ɛ]; you will once again notice the tongue lowering and retracting.

The final parameter along which English vowels can be described is whether they are produced with **lip rounding** or not. The rounded back vowels of English are [u, ʊ, o, ɔ, ɒ], whereas [ɑ] is unrounded. All the front and central vowels in English are unrounded. To see that ***rounding is a separate parameter from tongue height and backness***, try making an [i] sound while looking in the mirror. You will notice that the corners of the mouth are drawn back and that there is no lip rounding. Now without moving your tongue, round your lips by protruding them. The sound you wind up producing is a high tense front rounded vowel, a sound that does not occur in English but that is found in many languages of the world, such as German and French. As a point of interest, this sound is transcribed as [y] in the IPA. This is why we use [j] and not the [y] symbol to transcribe the palatal approximant found in English words like *yam* and *yellow*.

TEXTBOX 2.3 **VOWELS BEFORE "r" IN ENGLISH**

Combinations of vowels plus "ɾ" in English tend to be pronounced very differently from other vowel-plus-consonant sequences. Speakers of many dialects drop the "ɾ" entirely at the end of a word or syllable, as in the stereotypical pronunciation of the phrase [pak ðə ka ɪn ˈhavəd jad] 'Park the car in Harvard Yard' by speakers of Boston or British English. After certain vowels, as we have seen, the "ɾ" may leave a residual schwa-like vowel that combines with the preceding vowel to form a diphthong as in the words *peer, pear*, and *poor*, as produced by speakers of British English. The number of vowels occurring before "ɾ" (where it survives) is also typically reduced. Many speakers of American English thus have only one or two of the vowels [e, ɛ, æ] before "ɾ". The words *Mary, merry*, and *marry* are homophonous for many people, as are the names *Karen* and *Keren*. Likewise, the words *pour* and *poor* are identical for many speakers.

In addition to the English vowels in Tables 2.2 and 2.3, ***there are also combinations of vowels in English that function as a single unit in the sound system***. These vowel sequences are called ***diphthongs***, in contrast to ***monophthongs***, which are produced with a single articulatory configuration. Diphthongs may be regarded as the vocalic equivalent

to affricates, since they involve two phases. There are three clear diphthongs in American English. One is [aɪ], which starts off as [a] and then rapidly sequences into [ɪ]. The diphthong [aɪ] is found in many words in English including *write, lie*, and *mine*. The second diphthong is [aʊ], which is found in words like *cow, town*, and *bout*. Finally, the diphthong [ɔɪ] occurs in words like *boy, soy*, and *toil*. British English has a few additional diphthongs corresponding to sequences of vowel plus [ɹ] for most speakers of American English. These diphthongs, which all end in schwa, include [ɪə] in words like *peer* and *fear*, [eə] in words like *pear* and *care*, and [ʊə] in words like *poor* and *tour*. Textbox 2.4 describes some of the other diphthongs that occur in English, which are a little less clear.

TEXTBOX 2.4 **DIPHTHONGS IN BROAD AND NARROW PHONETIC TRANSCRIPTIONS**

Students with a careful ear will notice that most pronunciations of the English tense mid vowels [e] and [o] are actually diphthongs, with the tongue moving from the position of [e] to the position of [ɪ] in a word like *lay* or either [o] to [ʊ] (American English) or [ə] to [ʊ] (British English) in a word like *show*. A **broad phonetic transcription** would transcribe these words as [le] and [ʃo]. A **narrower phonetic transcription**, one which seeks to record as much detail as possible, would transcribe them as [leɪ] and either [ʃoʊ] or [ʃəʊ].

2.4 Using the IPA to Transcribe Words

Phonetic transcription exercise

Now that you are familiar with the IPA, we can try using it to transcribe English words. ***One of the most important aspects of doing phonetic transcription is not to be biased by the spelling of a word***. Since spelling symbols often differ from IPA symbols, as we have seen, blindly following the English orthography can lead to transcription mistakes. In doing transcription, you may find it useful to first think about how many sounds are in the word you are transcribing. To take a simple example, the English word *do* has two sounds. The first sound is a voiced alveolar stop [d], while the second sound is a high back rounded vowel [u]. (Don't be influenced by the spelling of the vowel as "o.") If we put the two sounds together, we get [du] as the phonetic transcription. Let's take a slightly trickier example now. In the spelling of the word *checks*, there are six letters. In terms of phonetic transcription, however, there are only four sounds. The first sound is the voiceless postalveolar affricate [tʃ]. The vowel is the lax mid-front vowel [ɛ]. The final consonant sequence consists of a voiceless velar stop [k] and a voiceless alveolar fricative [s]. (Note that these are only two sounds [k] and [s], even though there are three letters, "cks," in the spelling.) With a little practice, you should be able to phonetically transcribe any English word, or even words in other languages containing sounds that have been introduced in this chapter. It is also possible to go in reverse and sound out the word from the phonetic transcription. For

SIDEBAR 2.7

Your understanding of phonetics, phonology (the subject of the next chapter), and linguistics more generally will be easier if you spend time now memorizing the IPA: what each symbol means and how it is described. To help you with this, two sets of online flashcards are available on the website: one takes you from standard description to symbol and the other from symbol to description.

IPA flashcards